

MINERAL METABOLISM

Biochemistry Notes - calcium, phosphorus, magnesium, sodium, potassium, iron, copper, molybdenum, selenium and other important trace elements

What these notes cover

- Definition, general functions and classification of minerals.
- Major macrominerals: calcium, phosphorus, magnesium, sodium and potassium.
- Important trace elements: iron, copper, molybdenum, selenium and other high-yield elements.
- Functions, sources, key serum values and common deficiency/toxicity disorders.

Definition and general importance

Minerals are inorganic elements required for normal growth, maintenance and biochemical functioning of the body. They form only a small fraction of body weight, but they are essential for skeletal structure, fluid balance, enzyme activity, nerve and muscle function, hormone synthesis, oxygen transport and antioxidant protection.

Classification of minerals

Class	Requirement / body amount	Examples
Principal elements / macrominerals	Required in larger amounts, usually >100 mg/day; form most of the inorganic material of the body.	Calcium, phosphorus, magnesium, sodium, potassium, chloride, sulfur
Essential trace elements / microminerals	Required in very small amounts, usually <100 mg/day, but necessary for metabolism.	Iron, copper, iodine, manganese, zinc, molybdenum, cobalt, fluorine, selenium, chromium
Possibly essential / non-essential trace elements	Biological need is uncertain or not established; some may be toxic in excess.	Nickel, vanadium, cadmium, barium; non-essential examples include aluminium, lead, mercury, silver

General functions of minerals

- Bone and teeth mineralization: calcium, phosphorus, magnesium and fluoride strengthen skeletal tissue.
- Fluid, osmotic and acid-base balance: sodium, potassium and chloride regulate ECF/ICF balance and blood pH.
- Neuromuscular activity: calcium, magnesium, sodium and potassium maintain nerve impulse and muscle contraction.
- Enzyme cofactors: Mg, Mn, Zn, Cu, Mo, Se and K activate or form parts of enzymes.
- Special biological molecules: iron in hemoglobin, iodine in thyroid hormones, cobalt in vitamin B12 and zinc in insulin storage/secretion.

Principal minerals

Calcium (Ca)

Calcium is the most abundant mineral in the body; about 99% is present in bones and teeth as mineralized calcium phosphate/hydroxyapatite.

Sources / requirement	Serum / key value
Best sources: milk and milk products. Good sources: beans, leafy vegetables, fish, cabbage and egg yolk. Adult requirement: about 800 mg/day; higher in pregnancy, lactation and post-menopause.	Normal serum calcium: 9-11 mg/dl. About 50% is ionized, 40% protein-bound and 10% complexed with citrate/phosphate/bicarbonate.

Biochemical functions

- Forms and maintains bones and teeth along with phosphate.
- Triggers muscle contraction by interacting with troponin C and activating ATPase activity.
- Required in blood coagulation as factor IV.
- Necessary for nerve impulse transmission and membrane permeability.
- Acts as an intracellular second messenger and participates in hormone release, secretion, endocytosis and cell motility.
- Activates enzymes such as pancreatic lipase, ATPase and succinate dehydrogenase.

Deficiency / excess / disease states

- Hypocalcemia: serum Ca below about 7 mg/dl may cause tetany with neuromuscular irritability, spasms and convulsions.
- Hypercalcemia: seen in hyperparathyroidism and may cause increased urinary Ca/P loss and stone formation.
- Rickets/osteomalacia: defective bone calcification due to deficiency of calcium, phosphorus or vitamin D.
- Osteoporosis: demineralization and progressive loss of bone mass, common in elderly and post-menopausal women.

High-yield note

- Absorption is promoted by calcitriol, PTH, acidity, lactose, lysine and arginine; inhibited by phytate, oxalate, high phosphate, free fatty acids, alkaline pH and excess fiber.

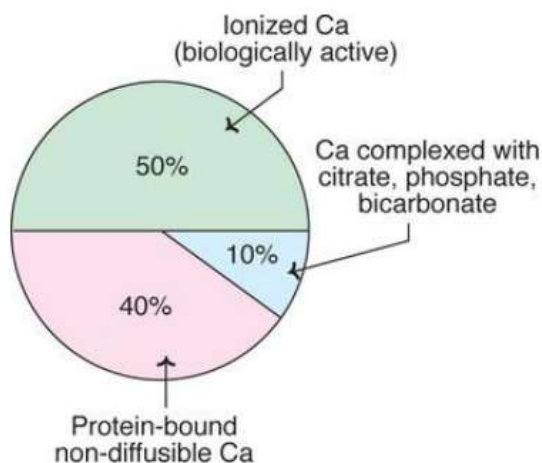


FIG. 18.1 Different forms of circulating calcium.

Source diagram: different forms of circulating calcium.

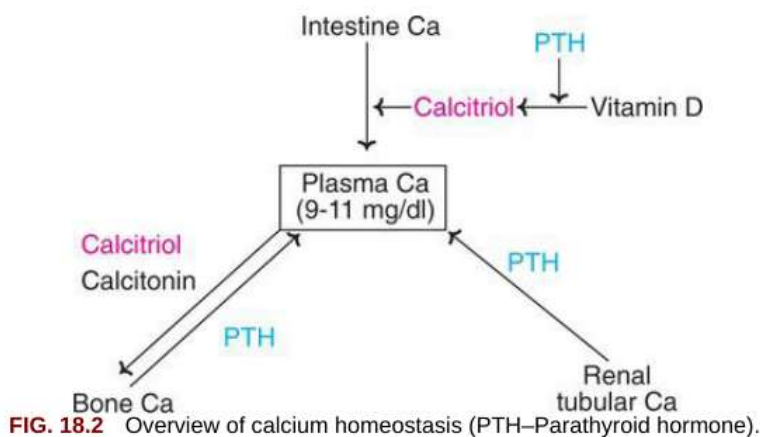


FIG. 18.2 Overview of calcium homeostasis (PTH–Parathyroid hormone).

Source diagram: overview of calcium homeostasis regulated by calcitriol, PTH and calcitonin.

Phosphorus (P / phosphate)

Phosphorus is closely associated with calcium and is essential in both structural and metabolic functions.

Sources / requirement	Serum / key value
Sources: milk, cereals, leafy vegetables, meat and eggs. Adult intake is usually linked with calcium; ideal Ca:P ratio is around 1:1, with absorption optimal between 1:2 and 2:1.	Serum phosphate: about 3-4 mg/dl; whole blood is higher because RBCs and WBCs contain much phosphate.

Biochemical functions

- Forms bones and teeth with calcium as calcium phosphate/hydroxyapatite.

- Central component of high-energy compounds such as ATP, GTP and creatine phosphate.
- Present in phospholipids, phosphoproteins and nucleic acids DNA and RNA.
- Component of nucleotide coenzymes such as NAD⁺, NADP⁺, ADP and AMP.
- Protein phosphorylation activates many enzymes and signaling proteins.
- Phosphate buffer system helps maintain blood and cellular pH.

Deficiency / excess / disease states

- Serum phosphate rises in hypoparathyroidism and falls in hyperparathyroidism.
- Severe renal disease may increase serum phosphate and contribute to acidosis.
- Vitamin D deficiency rickets and renal rickets are associated with low serum phosphate.

Magnesium (Mg)

Magnesium is present mainly in bones along with calcium and phosphorus; the rest is in soft tissues and body fluids.

Sources / requirement	Serum / key value
Sources: cereals, nuts, beans, vegetables, meat, milk and fruits. Requirement: about 350 mg/day in adult men and 300 mg/day in adult women.	Serum Mg: about 2-3 mg/dl; mostly ionized, with some complexed or protein-bound.

Biochemical functions

- Required for formation of bones and teeth.
- Cofactor for ATP-requiring enzymes such as hexokinase, glucokinase, phosphofructokinase and adenylate cyclase.
- Needed for normal neuromuscular function; low Mg increases neuromuscular irritability.
- Supports phosphorylation reactions and energy metabolism because Mg²⁺ complexes with ATP.

Deficiency / excess / disease states

- Deficiency causes neuromuscular irritation, weakness and convulsions resembling tetany.
- Malnutrition, alcoholism, cirrhosis, uremia, rickets and abnormal pregnancy may be associated with low Mg.

Sodium (Na)

Sodium is the chief extracellular cation and is closely related to water balance and blood pressure.

Sources / requirement	Serum / key value
Major source: common salt (NaCl). Other sources: bread, whole grains, leafy vegetables, nuts, eggs and milk. Requirement commonly 5-10 g NaCl/day; lower intake is advised in hypertension.	Serum sodium: 135-145 mEq/l. Aldosterone promotes renal sodium reabsorption.

Biochemical functions

- Regulates acid-base balance with chloride and bicarbonate.
- Maintains osmotic pressure and extracellular fluid volume.
- Required for normal muscle irritability and cell permeability.
- Participates in intestinal absorption of glucose, galactose and amino acids.
- Necessary for initiating and maintaining the heartbeat.

Deficiency / excess / disease states

- Hyponatremia: seen in diarrhea, vomiting, chronic renal disease, Addison disease and excess salt-free fluids; may cause low blood pressure and circulatory failure.
- Hypernatremia: seen in dehydration, Cushing syndrome and rapid sodium administration; may increase blood volume and blood pressure.
- Salt restriction is clinically important in hypertension, edema and congestive cardiac failure.

Potassium (K)

Potassium is the principal intracellular cation and is vital for neuromuscular and cardiac function.

Sources / requirement	Serum / key value
Sources: banana, orange, pineapple, potato, beans, chicken, liver and tender coconut water. Requirement: about 3-4 g/day.	Serum potassium: about 3.4-5.0 mEq/l. Hemolysis falsely raises measured serum K because RBCs contain high K.

Biochemical functions

- Maintains intracellular osmotic pressure.
- Regulates acid-base balance and water balance inside cells.

- Required for optimal activity of pyruvate kinase in glycolysis.
- Necessary for nerve impulse transmission and protein biosynthesis by ribosomes.
- Extracellular K⁺ strongly influences cardiac muscle activity.

Deficiency / excess / disease states

- Hypokalemia: caused by Cushing syndrome, cortisone therapy, K-free IV fluids, insulin treatment of diabetic coma, diarrhea and vomiting; causes weakness, tachycardia and ECG changes.
- Hyperkalemia: seen in renal failure, Addison disease, diabetic coma, severe dehydration or excess potassium infusion; may cause bradycardia, mental confusion and cardiac arrest.

Important trace elements

Iron (Fe)

Iron is essential for oxygen transport, cellular respiration and heme-containing enzymes. Adult body iron is about 3-5 g; nearly 70% is in hemoglobin.

Sources / requirement	Serum / key value
Sources: liver, meat, egg yolk, whole grains, cereals and green leafy vegetables. Absorption is promoted by gastric acid, ascorbic acid, cysteine, fructose and small peptides; inhibited by phytate, oxalate and high phosphate.	Plasma iron is bound to transferrin. About 1-2 mg is absorbed per day to replace normal loss; iron is efficiently reutilized.

Biochemical functions

- Forms hemoglobin and myoglobin for oxygen transport and storage.
- Part of cytochromes and iron-sulfur proteins involved in electron transport and oxidation-reduction reactions.
- Required for enzymes such as catalase, peroxidase and cytochrome oxidase.
- Stored mainly as ferritin and hemosiderin; transported in plasma by transferrin.

Deficiency / excess / disease states

- Iron deficiency anemia: the most prevalent nutritional disorder; classically microcytic hypochromic anemia.
- Causes include inadequate intake, poor absorption, chronic blood loss, pregnancy, lactation and growth.
- Excess iron causes hemosiderosis due to tissue deposition of hemosiderin; severe overload may damage organs.

High-yield note

- Fe³⁺ is reduced to Fe²⁺ for better absorption. Ceruloplasmin, a copper protein, helps oxidize Fe²⁺ to Fe³⁺ for transferrin binding.

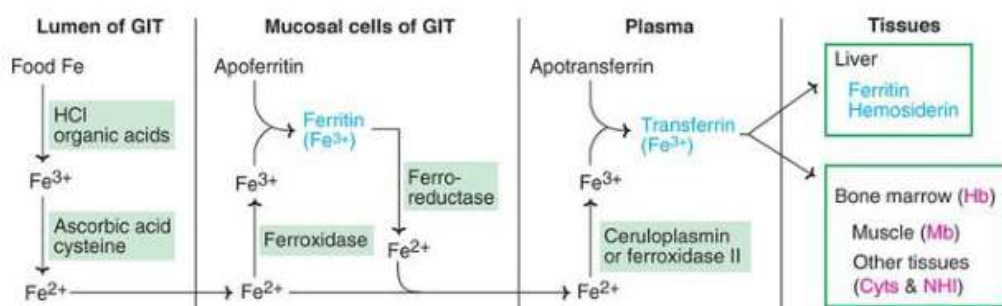
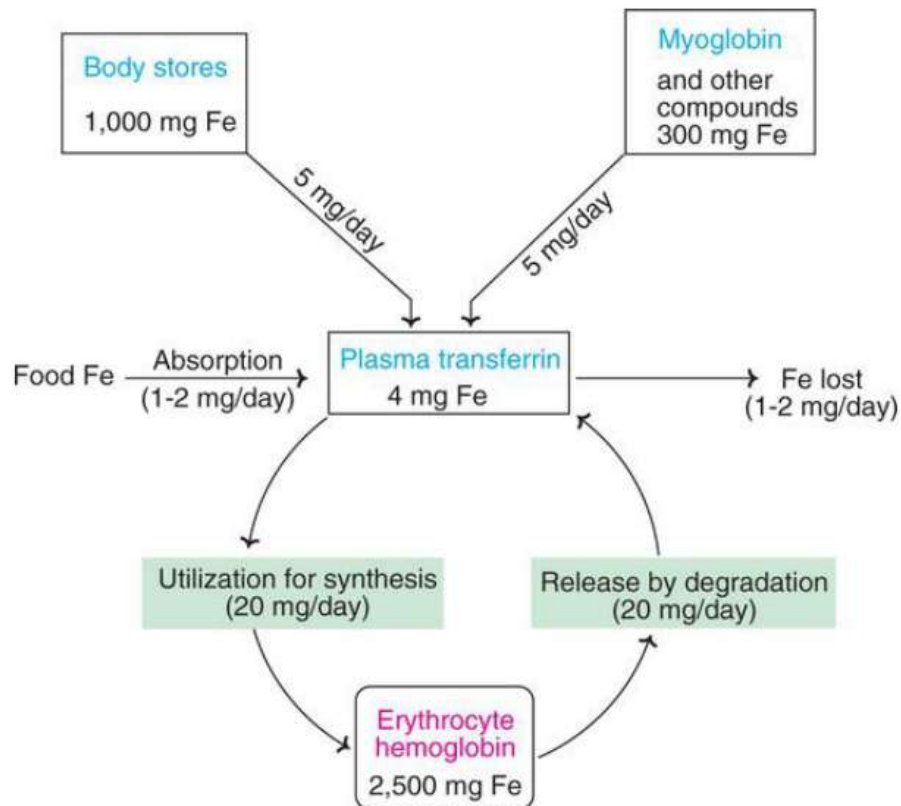


FIG. 18.3 Iron absorption and transport (GIT–Gastrointestinal tract; Hb–Hemoglobin; Mb–Myoglobin; Cyts–Cytochromes; NHI–Non-heme iron).

Source diagram: iron absorption and transport through intestinal mucosa, plasma and tissues.



Source diagram: general overview of iron metabolism and reutilization.

Copper (Cu)

Copper is required in small amounts and is part of several oxidase enzymes and iron metabolism proteins.

Sources / requirement	Serum / key value
Sources: liver, eggs, fish, oysters, nuts, seeds and vegetables. Most plasma copper is associated with ceruloplasmin.	Copper circulates mainly as ceruloplasmin; deficiency lowers enzyme activity and affects iron metabolism.

Biochemical functions

- Constituent of enzymes such as cytochrome oxidase, catalase, tyrosinase and lysyl oxidase.
- Ceruloplasmin transports copper and has ferroxidase activity needed for iron transport.
- Supports melanin synthesis through tyrosinase and connective tissue cross-linking through lysyl oxidase.
- Participates in hemoglobin formation indirectly by helping iron mobilization and transport.

Deficiency / excess / disease states

- Deficiency may cause anemia, bone demineralization, demyelination, arterial fragility, hypopigmentation and growth defects.
- Wilson disease: abnormal copper metabolism with deposition in liver, brain and kidney; plasma copper and ceruloplasmin are usually low.
- Menkes disease: inherited defect of copper transport causing severe deficiency features.

Molybdenum (Mo)

Molybdenum is a trace element mainly important because it forms part of specific enzymes.

Sources / requirement	Serum / key value
Widely distributed in natural foods. Dietary Mo is effectively absorbed by the small intestine.	Requirement is not clearly established in the source; only small trace amounts are needed.

Biochemical functions

- Constituent of xanthine oxidase, aldehyde oxidase and sulfite oxidase.
- Involved in purine catabolism, aldehyde oxidation and sulfite metabolism.
- Plant nitrite reductase contains molybdenum and is important in nitrogen metabolism.

Deficiency / excess / disease states

- Molybdenosis is a rare toxicity due to excessive intake; it may cause impaired growth, diarrhea and anemia.
- Excess Mo can diminish intestinal copper absorption and reduce copper utilization.

Selenium (Se)

Selenium is biologically important in antioxidant defense and thyroid hormone metabolism.

Sources / requirement	Serum / key value
Sources: organ meats such as liver and kidney, and sea foods. Recommended adult intake in the source: 50-200 micrograms/day.	Low serum selenium has been associated in epidemiological studies with higher risk of cardiovascular disease and some cancers.

Biochemical functions

- Works with vitamin E to protect against hepatic necrosis and muscular dystrophy.
- Maintains structural integrity of biological membranes.
- Present as selenocysteine in glutathione peroxidase, protecting cells from H₂O₂ and lipid peroxidation.
- 5-deiodinase, a selenium enzyme, converts thyroxine (T₄) to triiodothyronine (T₃).
- Thioredoxin reductase is also a selenoprotein and participates in nucleotide metabolism.
- Binds heavy metals such as mercury and cadmium, reducing toxicity.

Deficiency / excess / disease states

- Deficiency in animals causes muscular dystrophy, pancreatic fibrosis and reproductive disorders.
- Keshan disease: endemic cardiomyopathy associated with Se deficiency.
- Selenosis: toxicity causing weight loss, emotional disturbances, diarrhea, hair loss and garlic odor in breath.

Other important minerals - quick revision table

Mineral	Main biochemical functions	Sources / key point	Deficiency / excess
Chloride (Cl)	Acid-base balance, fluid/osmotic pressure, gastric HCl formation, chloride shift, activates salivary amylase.	Common salt, grains, leafy vegetables, eggs, milk.	Hypochloremia: vomiting, diarrhea, alkalosis, Addison disease; hyperchloremia: renal disease/dehydration.
Sulfur (S)	Present in cysteine, cystine, methionine, thiamine, biotin, lipoic acid and coenzyme A; detoxification via sulfate conjugation.	Protein-rich foods.	Rare as isolated deficiency; linked to protein/amino acid nutrition.
Iodine (I)	Required for synthesis of thyroid hormones T ₃ and T ₄ .	Iodized salt, sea foods.	Deficiency causes endemic goiter and hypothyroid features; excess may disturb thyroid function.
Manganese (Mn)	Cofactor for several enzymes; involved in bone formation, metabolism and antioxidant enzyme systems.	Cereals, nuts, tea, vegetables.	Deficiency in animals: poor growth, bone deformities, sterility, fatty liver; excess may affect CNS.
Zinc (Zn)	Component of carbonic anhydrase, alcohol dehydrogenase, alkaline phosphatase, carboxypeptidase and superoxide dismutase; required for insulin storage/secretion, vitamin A transport, wound healing, taste and reproduction.	Meat, fish, eggs, milk, beans, nuts.	Deficiency: growth retardation, poor wound healing, anemia, loss of appetite/taste, impaired spermatogenesis; acrodermatitis enteropathica.
Cobalt (Co)	Constituent of vitamin B ₁₂ ; supports erythropoiesis via B ₁₂ -dependent functions.	Foods containing vitamin B ₁₂ .	Toxic prolonged administration may cause polycythemia.
Fluorine (F)	Prevents dental caries by forming fluoroapatite; supports bones; inhibits enzymes like enolase at high levels.	Drinking water is main source.	Low fluoride: dental caries; excess: dental fluorosis, skeletal fluorosis, genu valgum.
Chromium (Cr)	Promotes insulin action and glucose utilization; lowers serum cholesterol; influences lipoprotein metabolism.	Brewer yeast, grains, cereals, cheese, meat.	Deficiency: disturbed carbohydrate/lipid/protein metabolism; excess may damage liver and kidney.

High-yield serum values and one-line memory points

Parameter	Normal / key value	Exam memory point
Serum calcium	9-11 mg/dl	Regulated by calcitriol, PTH and calcitonin.
Serum phosphate	3-4 mg/dl	High in hypoparathyroidism; low in

		hyperparathyroidism/rickets.
Serum magnesium	2-3 mg/dl	Many ATP-dependent kinases require Mg ²⁺ .
Serum sodium	135-145 mEq/l	Chief extracellular cation.
Serum potassium	3.4-5.0 mEq/l	Chief intracellular cation; dangerous for heart if too high or low.
Iron storage	Ferritin and hemosiderin	Transferrin transports Fe ³⁺ in plasma.
Copper protein	Ceruloplasmin	Has ferroxidase activity for iron transport.
Selenium enzyme	Glutathione peroxidase	Antioxidant protection against H ₂ O ₂ /lipid peroxides.

Final exam tip

- For long answers, write each mineral under: body distribution, functions, sources/requirement, absorption or transport, serum value and disorders.
- Calcium and iron are the most diagram-friendly answers: add calcium homeostasis and iron metabolism diagrams for extra marks.
- For electrolyte questions, contrast sodium as extracellular and potassium as intracellular.